

Technical Appendix: An Empirical Analysis of Transfer for Dynamic Path Planning

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Abstract

Complete specifications, per-suite tables, classical-control and replication detail, audit trails, and the artifact inventory for the main paper.

This appendix is organized by claim: exact specifications (A–B), the dynamic zero-shot result with its replications, classical controls, and audit trail (C), adaptation and the matched-label factorial (D), boundary experiments (E–H), validity forensics (I), statistics and the map-clustered reanalysis (J), and reproducibility (K).

A. Protocols by Stage

All continuous stages share the substrate: a 2-D point robot, maps with procedurally generated obstacles, a probabilistic roadmap (PRM) of exactly 192 nodes with $k=7$ connectivity, and A^* evaluated at a per-suite *binding budget* of node expansions calibrated (on the anchor baseline only) toward pre-specified target success rates; the exact grid, selection rule, and realized rates are in Table 5. Per-stage protocol essentials are summarized in Table 1. Unless noted, training labels are exact graph-Dijkstra costs on the PRM (static) or exact backward space–time Dijkstra values over (node, t) (dynamic). Headline continuous runs are local single-GPU (RTX 5090) validations with one primary training seed unless stated; the discrete program ran on a remote fleet.

B. Exact Environment, Model, and Data Specifications

World generation. Continuous worlds are unit squares (side length 1.0; the large-room family uses 2.0); units are abstract. Start and goal are sampled with minimum separation 0.45 of the side length. Each roadmap has 192 nodes in total—the start and goal at indices 0 and 1 plus 190 sampled free-space nodes—with $k=7$ nearest-neighbor candidate edges per node (doubled for the start and goal, then symmetrized); edges are validated by collision sampling at resolution 0.010 of the side length, and worlds whose roadmap fails to connect start to goal are rejected and resampled.

Dynamic data collection. A candidate training world is *usable* iff (i) the world builds, (ii) its roadmap connects start to goal, and (iii) the space–time problem is solvable from the start at $t=0$ (finite backward space–time Dijkstra

value). Training worlds are drawn only from the dynamic maze, rooms, and spiral families. Collection draws a fixed number of candidate worlds per family from a deterministic per-(family, index) seed stream and keeps the usable ones: the canonical run drew 24 per family and kept 53 (maze 17 / rooms 19 / spiral 17); the seed-2001 and seed-2002 retrains drew 64 per family and kept 139 (41/50/48) and 149 (40/54/55)—the collection-budget divergence disclosed in Section C. Every usable world has exactly 192 nodes and $t_{\max}=110$, i.e. 21,312 node–time slots; the training loss is masked to states with finite backward space–time values. Exact supervised-label counts from the deterministic recount (which reproduces each run’s per-suite world counts as a correctness gate): canonical 821,850 of 1,129,536 slots (72.8%; per-world mean 15,507, range 13,028–17,322); seed 2001, 2,115,331 of 2,962,368 (71.4%); seed 2002, 2,257,512 of 3,175,488 (71.1%). The reachable fraction is family-dependent (the maze-dense target worlds of the factorial average $\sim 18,900$ usable states per world).

C. Zero-Shot Transfer: Tables, Replications, Weighted A^* , and Audit Trail

Tables 6 and 10 give the complete per-suite C7 static and C8 dynamic results summarized in the main paper; Tables 7–9 give the fixed-provider primary view and its replications.

Multi-seed replication detail (Table 9). Two full-pipeline retrains (collect + train, seeds 2001/2002) of the aware/blind field U-Net twins were run under a design frozen before training, then evaluated on the same frozen fresh cohort with the canonical binding budgets. *Disclosed protocol divergence:* the replication design specified collection “identical to canonical” but set 64 candidate worlds per family, whereas the canonical run had used 24; the retrains therefore trained on 139 and 149 usable worlds versus the canonical 53 (per-family counts in Section B). The success replication is thus a replication under independent seeds *and* a larger training collection; the two retrains are collection-matched with each other. This divergence was discovered post hoc during the label-count audit and is reported rather than repaired. Matched-ratio medians across seeds: maze 0.047–0.059, rooms 0.117–0.135, spiral 0.065–0.092 (seed-stable); crossing 0.273–0.434, dense maze 0.216–0.625, large rooms 0.250–0.548 (seed-variable).

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Stage	Protocol essentials
C1–C4	9 easy suites, $n=80/40$ maps, budgets 100–200; pooled bases, TaskLoRA experts, RBF mixtures.
C5	calibrated hard maze/dense/rooms; 160 train maps; 80 eval maps attempted per suite, 80/79/77 retained after world-build filters; budgets 128–168; scalar residual target with cap.
C6	goal-conditioned raster residual field (8 input channels), sampled at PRM nodes; 96/40 maps; budgets 128–168.
C7	6 suites (3 held-out families); 96/24 maps per suite; binding budgets 140/24/56; scalar+field $\times$ additive+focal matrix; exact-Dijkstra oracle.
C8	deterministic moving circular obstacles; space–time PRM; aware ($W=8$ future window) vs. blind ($W=0$) twins; canonical run: 53 usable training worlds (maze/rooms/spiral; collection details and exact label counts in Section B); 20-map/suite <i>development</i> cohort, then a 50-map/suite <i>confirmation</i> cohort evaluated after freezing, plus two independent full-pipeline retrains (139/149 worlds) on the same confirmation cohort.
C9/C9 _H	adapt C7 pooled bases to 3 held-out families; $K \in \{0, 1, 2, 4, 8, 16, 32\}$ labeled maps; LoRA(r8, bounded/unbounded), full FT, scratch; C9 _H matches compute (10 epochs, LR 2×10^{-4}); 30 fixed test maps/target.
C9 _B	same protocol on C8 dynamic sources (aware+blind), 486 adapters, 3 targets, $K \in \{1, 4, 16\}$.
C10	16 rank-8 experts on 2 family axes; zero-label transfer to 3 interior targets; RBF/nearest/uniform weight-merge and prediction-mix composition.
C11	compositional missions (waypoint sequences; key/door), configs A/B/C, mission length $K \in \{0, 2, 4, 8\}$; 40 train / 25 test maps per cell; 6 learned arms $\times$ 3 seeds (198 checkpoints; 54,450 eval rows); exact mission oracle and leg-sum baseline; forced-compute and would-halt probes.
C12	(A) hidden slow/fast regime dynamics, memory-headroom gate, matched temporal-hierarchy pilot; (B) tied vs. untied product-graph refiner, cycles 1–8, configs A/C $\times$ $K \in \{2, 8\}$, 3 seeds, 48 checkpoints, two 3,200-row test tables.
C13	current-state heuristic under bounded observation (radius 0.20 local subgraph); behavior-derived fresh-start rollout labels (median of 3); staged falsification ladder (A–L) then frozen 144-map confirmation (M); architecture substitutions (N/O); persistent-state pilot (P).
Discrete	20 $\times$ 20 to 256 $\times$ 256 grids; Manhattan-baseline A [*] ; forecasting (dynamic obstacles), additive residual transfer (22 suites $\times$ 3 budgets $\times$ 100 episodes), pooled multitask bases, learned focal ranking at $w \in \{1.0, 1.05, 1.1\}$.

Table 1: Protocol essentials by stage.

Family	Generator rule (lengths as fractions of side length)
Maze / dense maze	Corridor maze of axis-aligned walls; the dense variant narrows gap width to 0.115. Held out: dense.
Rooms / large rooms	2 $\times$ 2 rooms from one vertical + one horizontal partition wall, one doorway per wall, doorways biased toward opposite edges; large variant doubles side length. Held out: large.
Spiral	Four serpentine vertical walls spanning $x \in [0.22, 0.78]$, gap side alternating per wall. Trained.
Bugtrap	Solid vertical back wall, two horizontal arms, and a mouth lip closing the center of the opening (U-pocket). Held out.
Crossing (dyn.)	Open arena; dynamic timing control.

Table 2: Static geometry per family. Base-anchor families (open/clutter/narrow variants) used by the early stages additionally place 3–68 circular obstacles with radii 0.022–0.115; full ranges ship with the generator in the artifact package.

Preregistered readouts: R1 (success CI excludes zero in $\geq 5/6$ suites) 5/6 and 6/6—pass; R2 (deltas within ± 0.15 of canonical in ≥ 5 suites) 5/6 and 6/6—pass; R3 (no suite significantly aware-over-blind) fails as stated for seed 2001 (large rooms +0.080 [+0.020, +0.160])—the same suite that is significantly *blind*-better under seed 1234 (−0.200 [−0.320, −0.100]). Across the 18 seed–suite twin contrasts:

Suite	Patr.	Radius	Span	Period	v_{ag}	t_{max}
Maze	3	0.075	0.38	0.14	0.060	110
Rooms	3	0.075	0.40	0.14	0.060	110
Spiral	3	0.075	0.38	0.14	0.060	110
Dense maze	4	0.062	0.42	0.17	0.060	140
Crossing	4	0.055	0.50	0.30	0.060	110
Large rooms	5	0.075	0.36	0.14	0.120	130

Table 3: Dynamic-suite parameters. Each patroller is a circle sweeping a line segment with triangle-wave motion; radius and span are fractions of the side length, period is a fraction of the horizon $t_{max} dt$ ($dt=1$ s), and each patroller crosses its span twice per period. Sweep centers carry lateral jitter ≤ 0.02 (0.10 on crossing). The planner and collision checker always use these exact trajectories; only the *aware* heuristic input sees the $W=8$ future-occupancy channels.

2 significantly aware, 2 significantly blind, 14 null, with the significant cells in different suites for different seeds; the per-suite future-window effects flip sign across seeds and are consistent with training-seed noise. No retraining, reselection, or recalibration occurred in response to any readout.

Success as a function of budget (Figure 1). A single evaluation pass at $4\times$ the binding budget on the same frozen cohort yields *exact* success-vs-budget curves for every lower budget: the space–time A^{*} expansion order is budget-independent, and a map is solved at budget B iff its recorded solve expansions are $\leq B$, so the curves are derived by

Model	Configuration	Input	Parameters
Scalar HRM	hid. 192, 2 layers, 4 heads, $k=2$	62 tok. $\times$ 16	2,158,529
Scalar ON-LSTM	hid. 256, 2 layers, chunk 8	62 tok. $\times$ 16	1,252,481
Field U-Net	FiLM-cond.; base 32, enc. 32-256	64×64 raster	3,702,370
Field HRM	conv stem /4; hid. 256, 2 lay.	64×64 raster	4,016,642
Field ON-LSTM	conv stem /4; hid. 256, chunk 8	64×64 raster	1,448,578
Mission MLP	3 GELU layers, width 768	mission tok.	0.5–3.5M
Mission U-Net	FiLM stage emb. (32) at bottleneck	raster+stage	0.5–3.5M
Mission GNN	hid. 128, 8 untied rounds	graph	0.5–3.5M
Mission HRM-tr.	hid. 224, $k=2$, 4 heads, head 256	mission tok.	~ 2.90 M
Mission ON-L.-tr.	hid. 256, chunk 8, head 256	mission tok.	~ 1.25 M
Big U-Net / HRM	base 64 / hid. 512 (addendum)	as above	14.8M / 33.3M

Table 4: Instantiated architectures with parameter counts computed from the released constructors (mission arms are constructor-verified to lie in the preregistered [0.5M, 3.5M] band; the test suite asserts the counts). Scalar token sequences: one summary token, 12 nearest-obstacle tokens, 48 ray tokens, one goal token. Dynamic variants append W future-occupancy channels ($W=8$ aware; $W=0$ blind). LoRA inserts rank-8 adapters on the HRM (rank-24 on the ON-LSTM, matching its wider hidden layer); the quoted crossover cells are HRM (rank 8); the factorial’s dynamic arm applies rank-8 conv-LoRA to every U-Net Conv2d (192,080 trainable parameters). Optimizer for all models: AdamW, learning rate 2×10^{-4} (1.5×10^{-4} for adapters), weight decay 10^{-4} , batch 256, gradient clip 1.0, smooth-L1 loss on the capped residual (cap 4.0), 10 base epochs (8 adapter epochs) unless a stage states otherwise. Remaining constructor details (activations, normalization, heads) ship with the code in the artifact archive.

Suite	Band (binding)	A@cal.	O@cal.	A@conf.
Crossing	150/250	0.30	1.00	0.12
Maze	1800/2500	0.30	1.00	0.12
Dense maze	2500/3500	0.05	0.95	0.06
Rooms	1300/1800	0.30	1.00	0.42
Large rooms	600/900	0.75	1.00	0.82
Spiral	2500/3500	0.20	0.95	0.16

Table 5: Dynamic budget calibration. Grid: {150, 250, 400, 600, 900, 1300, 1800, 2500, 3500} expansions. Rule: for each of two targets evenly spaced in [0.45, 0.70], pick the grid budget with the closest anchor success on the 20 calibration worlds (ties smaller); the smaller selected budget is *binding* (listed first). A@cal./O@cal.: anchor/oracle success at binding on the calibration worlds—the coarse grid can land outside the target band (dense maze has no point near 0.45). A@conf.: anchor success on the 50-map confirmation cohort (cohort drift). Static stages use the same rule class on their own grids (binding 128–168; realized 0.25–0.62).

thresholding with no re-runs and no interpolation (design frozen before launch). As a determinism cross-check, thresholding at the binding budget reproduces every fresh-cohort success value exactly (all suites, all providers). The headline result is not an artifact of the calibrated operating point: Euclidean search needs $1.7\text{--}2.8\times$ the binding budget to reach the fixed blind provider’s binding-budget success (crossing $2.49\times$, maze $2.29\times$, dense maze $1.95\times$, rooms $1.73\times$, large rooms $1.94\times$, spiral $2.80\times$), and by $4\times$ the binding budget every provider converges to the same feasibility ceiling (0.94–1.00; on dense maze the 0.94 ceiling binds even the oracle, reflecting maps unsolvable within the time horizon).

Tuned weighted-A* control. The control was designed

Suite	Euclid succ.	Field-HRM succ.	Ratio [95% CI]
Maze	0.583	1.000	0.521 [0.450, 0.627]
Maze-dense	0.250	0.958	0.804 [0.707, 0.829]
Rooms	0.375	0.958	0.829 [0.775, 0.885]
Spiral	0.250	0.917	0.850 [0.742, 0.919]
Bugtrap	0.458	0.750	0.714 [0.533, 0.800]
Rooms-large	0.417	0.750	0.839 [0.646, 1.222]

Table 6: C7 static suites at binding budgets: success and matched-solved median expansion ratio (field HRM vs. Euclid). Success gains are BH-significant in 4/6 suites (maze, dense, rooms, spiral); bugtrap and rooms-large reach corrected significance only for other learned arms. Scalar-HRM ratios span 0.427–0.822. Matched n ranges 6–14.

Suite	Euclid	Blind U-Net	Ratio [95% CI]	n
Crossing	0.30	0.95	0.191 [0.166, 0.878]	6
Maze	0.30	1.00	0.093 [0.057, 0.137]	6
Dense maze	0.05	0.75	0.246 (descriptive)	1
Rooms	0.30	1.00	0.099 [0.050, 0.214]	6
Large rooms	0.75	0.95	0.228 [0.185, 0.418]	15
Spiral	0.20	0.95	0.087 [0.061, 0.287]	4

Table 7: C8 fixed-provider result on the *original* 20-map cohort (field U-Net blind; no future-motion input; additive mode; binding budgets; map-level bootstraps). Columns 2–3: success. Success deltas are $+0.20$ to $+0.75$ with all six paired CIs excluding zero.

and frozen *after* the learned confirmation results were known (its purpose is to answer the “weak anchor” objection those results raise); its tuning touched only the development cohort, and the tuned weights were evaluated exactly once on

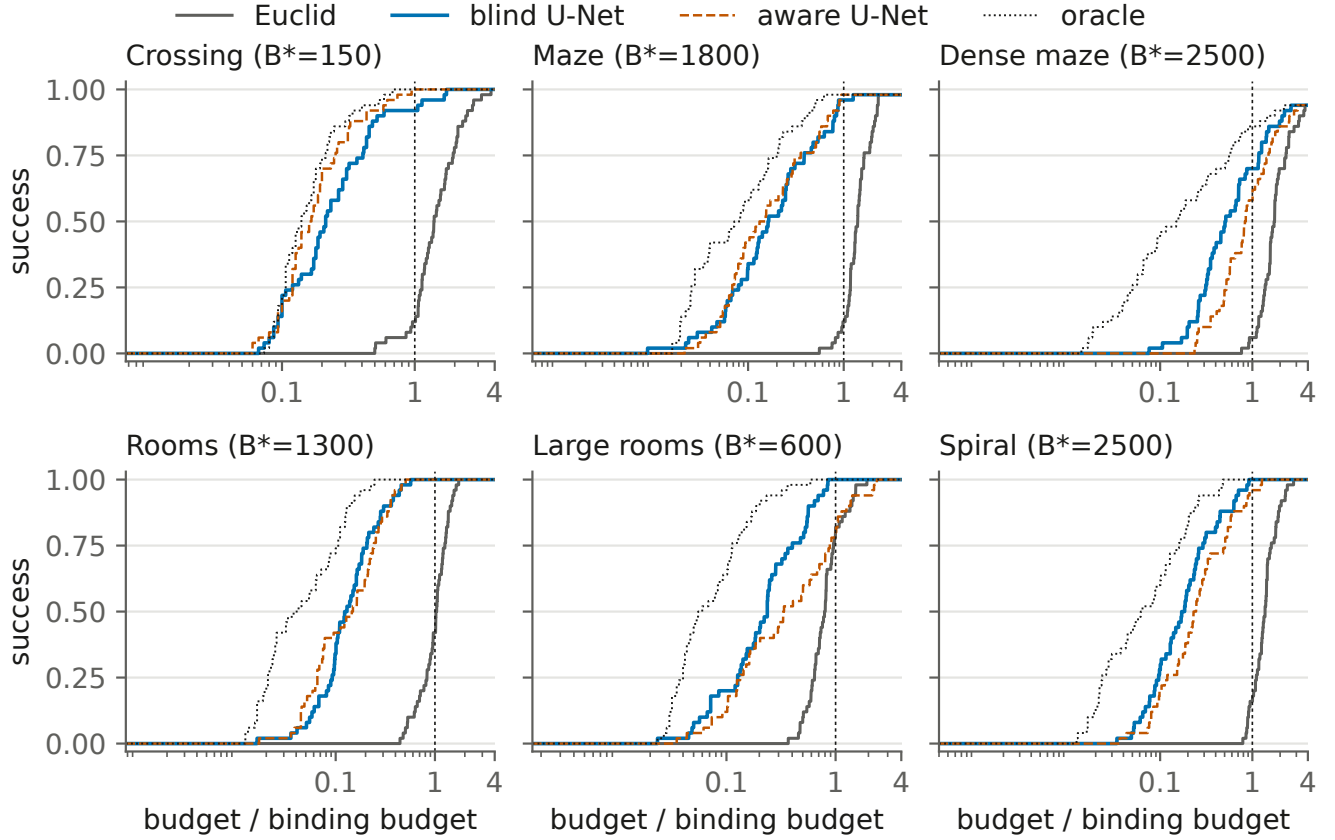


Figure 1: Success-vs-budget step curves per dynamic suite (fresh 50-map cohort), derived exactly from one 4 $\times$ -binding evaluation pass. Budgets are normalized by the binding budget B^* (dashed vertical line). The learned providers reach high success at $0.05\text{--}0.3 \times B^*$; Euclid requires $1.7\text{--}2.8 \times B^*$ to match the blind provider’s binding-budget success.

Suite	Euclid	Blind U-Net	Ratio [95% CI]	n
Crossing	0.12	0.92	0.273 [0.109, 0.516]	6
Maze	0.12	0.96	0.047 [0.021, 0.153]	6
Dense maze	0.06	0.70	0.216 [0.100, 0.281]	3
Rooms	0.42	1.00	0.121 [0.106, 0.172]	21
Large rooms	0.82	1.00	0.250 [0.188, 0.306]	41
Spiral	0.16	1.00	0.092 [0.065, 0.146]	8

Table 8: C8 fixed-provider *replication* on the fresh 50-map-per-suite cohort (generation seed 999999, frozen before any result was observed; map-disjoint from the original cohort by the evaluation seed formula’s within-suite offset bound). Success deltas are +0.18 to +0.84 with all six paired CIs excluding zero. Map-paired aware–blind success deltas on this cohort: crossing +0.080 [+0.020, +0.160], maze +0.020 [0.000, +0.060], dense maze -0.120 [-0.220 , -0.040], rooms 0.000, large rooms -0.200 [-0.320 , -0.100], spiral -0.040 [-0.100 , 0.000]: one suite significantly favors the aware twin, two the blind twin, three neither.

the confirmation cohort under the frozen rule (candidate grid $w_h \in \{1.1, 1.2, 1.5, 2, 3, 5\}$; highest development success, ties toward smaller w_h ; the anchor itself is the separately

reported $w_h=1$ arm). Table 11 reports the per-suite comparison with the exact McNemar policy applied: the only success difference that survives BH correction is spiral (15/0 discordant maps, $q=3.7 \times 10^{-4}$, favoring the learned heuristic); crossing’s four-map weighted-A* advantage does not (0/4 discordant, $q=0.375$), and the remaining suites have at most one discordant map. Median learned-to-weighted expansion ratios exclude parity on four of six suites; the crossing and dense-maze intervals include it.

Extended-grid check (frozen design 2026-07-26; specified post hoc). Because maze, dense maze, and spiral selected the grid maximum $w_h=5$, the grid was extended to $w_h \in \{7, 10\}$: the two new weights were run on the development cohort only, the identical frozen selection rule was re-applied over the merged grid, and only suites whose selection changed were evaluated once on the frozen confirmation cohort. Five selections are unchanged—in particular spiral’s development success is 0.80 at $w_h=5$ and no larger weight beats it, so its confirmation advantage (15/0 discordant) is grid-robust—while dense maze re-selects $w_h=7$ (development success 20/20 versus 18/20 at $w_h=5$; the development curve is monotone: 1/2/2/4/12/18/20/20 successes of 20 across the eight weights). At $w_h=7$ on the confirmation cohort, weighted A* solves 45/50 versus 35/50 for both

Suite	Seed 1234	Seed 2001	Seed 2002
Crossing	+0.80 [0.68, 0.90]	+0.86 [0.74, 0.96]	+0.88 [0.78, 0.96]
Maze	+0.84 [0.74, 0.94]	+0.86 [0.76, 0.96]	+0.84 [0.74, 0.94]
Dense maze	+0.64 [0.50, 0.78]	+0.46 [0.32, 0.60]	+0.56 [0.42, 0.70]
Rooms	+0.58 [0.44, 0.72]	+0.58 [0.44, 0.72]	+0.58 [0.44, 0.72]
Large rooms	+0.18 [0.08, 0.30]	+0.10 [−0.02, 0.22]	+0.16 [0.06, 0.26]
Spiral	+0.84 [0.74, 0.94]	+0.84 [0.74, 0.94]	+0.84 [0.74, 0.94]

Table 9: C8 preregistered multi-pipeline replication: paired success deltas (blind U-Net — Euclid, map-level 95% CIs) for three independently trained pipelines on the same frozen fresh cohort (the retrains used larger training collections; see the multi-seed detail paragraph). 17 of 18 CIs exclude zero; the one interval that includes zero is seed 2001’s near-ceiling large-rooms cell (Euclid 0.82).

Suite (arm)	Succ. E→L	Ratio [95% CI]	n
Crossing (U-Net)	0.30→1.00	0.258 [0.132, 0.769]	6
Maze (U-Net)	0.30→1.00	0.064 [0.046, 0.088]	6
Maze-dense (U-Net)	0.05→0.75	0.278 (no CI)	1
Rooms (U-Net)	0.30→1.00	0.096 [0.038, 0.191]	6
Rooms-large (U-Net)	0.75→0.90	0.380 [0.132, 0.591]	14
Spiral (field HRM)	0.20→0.90	0.046 [0.038, 0.073]	4

Table 10: C8 *secondary* view: per-suite strongest arms (explicitly post-hoc selection). Corrected success evidence holds on five suites; rooms-large is supported mainly by expansion evidence. Dense-maze has matched $n=1$ and supports no expansion inference. Blind arms are the per-suite best in five of six suites.

the frozen $w_h=5$ rows and the learned heuristic (15/5 discordant in each pairing, exact $p=0.041$, single post-hoc comparisons), with matched-solved learned-to-weighted expansions at parity (median 1.172 [0.948, 1.479], $n=30$) and cheaper weighted search on jointly solved maps (mean 843 versus 1,217 expansions at $w_h=5$). The dense-maze effort comparison in Table 11 is therefore grid-limited: under stronger inflation the control overtakes the learned arm on this suite, and the learned advantage concentrates where no tested inflation closes the gap (spiral). Raw development/confirmation rows and the selection report ship in the artifact package.

Wall-time decomposition (dynamic arms). On a 5-map-per-suite CPU sample at binding budgets: building the learned heuristic table (raster/feature construction plus $t_{\max}+1$ U-Net forwards) averages 1.23 s per map (range 0.95–1.59 s; one forward is ~ 5.9 ms, so inference is roughly half), and the learned search itself 0.31 s—about 1.54 s end to end, versus 0.39 s for tuned weighted A^* and 0.76 s for the anchor. At this graph scale, fewer expansions do not translate into lower wall time for the learned method; the dynamic result is a search-effort and success result, not a latency result (single-process CPU; script and raw timings in the artifact package).

SIPP baseline (frozen design 2026-07-25). Discrete-time Safe Interval Path Planning on the identical substrate: per-node safe intervals are maximal runs of wait-safe steps under the exact wait predicate (plus pass-through singletons, matching the space–time graph, where arrival never requires a node check), successors take the earliest feasible departure per

(edge, target interval) under the exact edge predicate, and the heuristic is the anchor h_0 . *Correctness gate, pre-committed:* on every instance, SIPP’s earliest arrival must equal the backward space–time Dijkstra optimum; all 300 confirmation instances pass, and solvability agrees everywhere. Results are tabulated in Table 12: run unbudgeted, SIPP solves 0.98–1.00 per suite at optimal path quality, and its interval-state expansions (a different unit from (v, t) expansions; never merged) have small per-suite medians. Per the design’s pre-committed framing: the learned result is search-effort structure within the budgeted time-expanded formulation, not superiority over a dedicated dynamic planner on this deterministic substrate. Raw rows with per-instance gate checks and timings ship in the artifact package.

External benchmark: MovingAI-derived worlds (frozen design 2026-07-25). Six MovingAI maps (Sturtevant 2012)—street: Berlin_0_256, Boston_0_256, Paris_0_256; dungeon (dao): den312d, den520d, brc202d—are majority-coarsened to the canonical 64×64 occupancy over the unit square and rectangle-decomposed into standard obstacles; the trained maze-family patroller configuration is applied on top; start/goal sampling, usability rules, the 192-node roadmap, budget calibration (anchor-only, canonical rule on 10 development instances per group), and weighted- A^* tuning are all canonical and were frozen before execution. Evaluation: 25 instances per group, every arm run once at $4\times$ the largest grid budget with success-at-binding derived by expansion thresholding. *Instance accounting:* the usability rules retain evaluation instances unevenly across base maps—street: Berlin 10, Boston 6, Paris 9; dungeon: den312d 8, den520d 17, brc202d 0 (its coarsened geometry never yields a usable roadmap instance, so five of the six selected maps contribute). All group-level tests are instance-level and unadjusted, with instances nested in three (street) and two (dungeon) base maps—a clustering limitation we flag rather than model at this sample size. Results (binding street 600, $w_h=1.5$; dungeon 900, $w_h=2$): learned success 0.68 on both groups versus anchor 0.76/0.76 and tuned WA^* 0.76/0.96; learned–anchor deltas -0.08 $[-0.20, 0.00]$ (0/2 discordant, $p=0.50$) and -0.08 $[-0.28, +0.12]$ (2/4, $p=0.69$); learned– WA^* -0.08 (street, $p=0.50$) and -0.28 $[-0.48, -0.12]$ (dungeon, 0/7 discordant, exact $p=0.016$). Matched expansion ratios: learned/anchor 0.481 [0.283, 0.643] on street (the one surviving advantage) but 0.796 [0.261, 1.670] on

Suite	w_h	Disc. (q)	Bl./WA* [CI] (n)	WA*/anch. (n)	Sub. W/B
Crossing	1.5	0/4 (.375)	.825 [.53, 1.17] (46)	.139 (6)	1.008/1.164
Maze	5	1/1 (1.0)	.601 [.44, .84] (47)	.120 (6)	1.017/1.013
Dense mz.	5	1/1 (1.0)	.982 [.82, 1.09] (34)	.106 (3)	1.006/1.008
Rooms	3	1/0 (1.0)	.554 [.44, .80] (49)	.158 (21)	1.015/1.019
Large rms.	1.5	1/0 (1.0)	.728 [.54, .97] (49)	.335 (41)	1.003/1.083
Spiral	5	15/0 (3.7e-4)	.217 [.18, .31] (35)	.567 (8)	1.013/1.014

Table 11: Tuned weighted A* versus the fixed blind U-Net on the confirmation cohort at binding budgets (success rates in the main-paper table). w_h : per-suite inflation selected on the development cohort under the frozen rule. Disc.: discordant maps (learned-only / WA*-only) with the BH-adjusted exact McNemar q . Bl./WA*: median learned-to-weighted expansions on the n jointly solved maps with 10k map-bootstrap 95% CI, two decimals here (three-decimal values in the main-paper table and artifact package; below 1 = learned expands fewer). WA*/anch.: median weighted-to-anchor expansions on jointly solved maps (bootstrap CIs in the artifact package). Sub. W/B: mean arrival over optimal arrival, WA*/blind, computed on the same jointly solved maps as the main-paper table (paired blind-WA* differences with 95% CIs: crossing +.156 [+ .116, +.200]; maze -.005 [-.016, +.006]; dense maze +.002 [-.005, +.009]; rooms +.004 [-.008, +.017]; large rooms +.080 [+ .058, +.105]; spiral +.001 [-.007, +.010]). Paired success-delta bootstrap CIs, for continuity: crossing -.08 [-.16, -.02]; spiral +.30 [+ .18, +.42]; others within $\pm .02$ of zero.

Suite	SIPP / learned succ.	Med. int. exp.	t_{int}	t_{srch}
Crossing	1.00 / 0.92	76	1.44	0.14
Maze	0.98 / 0.96	253	1.14	0.24
Dense maze	0.98 / 0.70	316	1.38	0.23
Rooms	1.00 / 1.00	192	0.82	0.17
Large rooms	1.00 / 1.00	102	1.53	0.13
Spiral	0.98 / 1.00	329	0.83	0.20

Table 12: SIPP on the 50-map confirmation cohort. Success is unbudgeted (every solved instance verified optimal against the backward space-time Dijkstra value; 300/300 gate checks pass); learned success at binding budgets shown for context. Med. int. exp.: median interval-state expansions (SIPP’s own unit). $t_{\text{int}}/t_{\text{srch}}$: mean per-map interval-construction and search wall time (s).

dungeon; learned/WA* 1.050 [0.812, 1.218] street and 2.196 [1.254, 6.360] dungeon. Jointly solved suboptimality: learned pays +0.074 [+0.030, +0.136] (street) and +0.101 [+0.021, +0.217] (dungeon) over WA*. Verdict, reported as-is per the frozen design: the fixed heuristic’s success advantage does not transfer to externally authored geometry under this protocol; the zero-shot claims are bounded to the procedural substrate and its tested shifts. Raw rows, map manifest, and the conversion rule ship in the artifact package.

Fixed-model selection audit trail. The primary dynamic claim rests on one fixed learned heuristic; its provenance is as follows. (1) *Candidates*: the original dynamic run trained aware and blind twins of scalar HRM, scalar ON-LSTM, and field U-Net models. (2) *Selection*: the field U-Net blind twin was designated the single primary model for every suite during the fixed-model reanalysis design; the original 20-map evaluation results were known at that time, so the accurate statement is that the model, integration mode (additive), and per-suite binding budgets were **fixed before the 50-map confirmation cohort was generated**, not that selection predates all target outcomes. (3) *Freeze*: the confirmation

cohort’s generation seed (999,999) was chosen in advance; the evaluation-seed formula’s within-suite offset bound guarantees map-disjointness from every earlier cohort, and world fingerprints (start, goal, obstacle coordinates) were checked. (4) *Integrity*: the fixed checkpoint’s SHA-256 is b8378950545f8abdbf06d59568b5d8ab6069884c1c038d23a41a14b6dc17fb6f; the binding-budget calibration file is copied verbatim into every evaluation directory. (5) *Seeds*: the two additional training pipelines rerun collection and training from scratch with new seeds (2001, 2002), resampling training worlds; architecture, integration, binding budgets, and the evaluation cohort are identical, but the retrains used a larger collection budget than the canonical run (64 versus 24 candidate worlds per family; Section B), a divergence from the replication design’s “identical to canonical” intent that we disclose rather than suppress.

D. Adaptation: K -Indexed Transfer, Composition, and the Matched-Label Factorial

Grain note: the point values in this section are the original record-level curve summaries, retained for continuity with the run reports; the map-clustered reanalysis in Section J supplies the primary inferential values quoted in the main paper, and where the two differ (e.g., C9h exact-zero counts, C9b’s single positive cell) the map-level values govern.

Representative C9 HRM curves (ratio at success): maze-dense LoRA 0.650 (1.00) at $K=1$ to 0.730 (0.97) at $K=16$; maze-dense full FT 0.744 (0.76) to 0.571 (0.99) at $K=16$ and 0.552 at $K=32$; scratch 1.008 (0.37) to 0.808 (0.80); rooms-large full FT 1.128 (0.37) at $K=1$ to 0.500 (0.92) at $K=8$; bugtrap LoRA degrades from ≈ 0.696 (0.83) to 0.950 (0.48) at $K=32$ —LoRA is usually but not universally base-preserving. C9h (matched compute): bounded-minus-unbounded LoRA median ratio delta 0.000 ± 0.008 across 27 cells (22/27 exactly zero; max $|\Delta|$ 0.028); rooms-large field U-Net full FT reaches 0.404 at 0.97 success from 0.982 zero-shot while conv-LoRA stays at 0.992–1.002. C9b (dynamics): maze-dense aware U-Net zero-shot 0.145 (0.60), LoRA $K=16$ 0.059 (0.97), full FT $K=1$ 0.112 (0.70); aware-minus-blind success deltas at full-FT $K=16$ are 0.00 or -0.05 in all nine cells. C10: RBF-merge minus zero-shot expansion-ratio deltas by target/backbone: -0.022 , -0.012 , -0.024 , $+0.082$, $+0.012$, $+0.116$; weight-space vs. prediction-space deltas lie within $[-0.012, +0.007]$; own-axis RBF mass 0.986–0.998.

The matched-label factorial (C14)

A preregistered factorial (design and pre-execution amendment in the artifact package) tests whether the LoRA/full-fine-tuning contrast is governed by the number of supervised search states N , independent of domain and map count. Factors: $N \in \{256, 1024, 4096, 16384, 65536\}$ exact-count state samples $\times$ two coverage levels at fixed N (concentrated: the minimal world count $w_{\min}(N)$; distributed: $8 \times w_{\min}(N)$ worlds) $\times$ two domains $\times$ {unbounded rank-8 LoRA, full fine-tuning, scratch} $\times$ 3 seeds—180 adaptations, every cell at an identical 2,560 optimizer steps, evaluated on the frozen C9/C9b test cohorts at binding budgets. Sources are frozen: the C7 pooled scalar HRM base (static) and the C8 field U-Net blind twin (dynamic); the target is the maze-dense family in both domains. The measured $w_{\min}$ spans quantify the domains’ supervision densities: reaching $N=65,536$ takes 356 static worlds but 4 dynamic worlds (~ 185 vs. $\sim 18,900$ usable states per world).

Preregistered verdict: H-C14 rejected as stated. On matched-solved expansion ratios the hypothesized crossover point is degenerate: full fine-tuning is at or below LoRA at every tested N in all 12 (domain $\times$ coverage $\times$ seed) cells, and the preregistered regression’s full-FT $\times \log N$ interaction is null (-0.0025 [$-0.0087, +0.0028$]; Table 16). State count improves ratios ($\log_2 N$ coefficient -0.0140 [$-0.0190, -0.0081$]) but identically for both methods. Ratios condition on jointly solved maps, so collapsed-success cells can appear spuriously efficient; success is the factorial’s primary endpoint.

The preregistered coverage readout localizes the supported success effect. The low-supervision protection LoRA provides appears in *success*, and it is governed by distinct-world coverage at fixed N , not by state count (Table 15; main-paper factorial figure). Concentrated low-world cells degrade full fine-tuning and scratch in both domains (success vs. the frozen zero-shot source: -0.300 [$-0.550, -0.050$] in all three seeds at dynamic concentrated $N=256$ and $N=1,024$; -0.167 to -0.300 , all seeds significant, at static concentrated $N=256$), while the distributed cells at the same N rescue them (dynamic $+0.150$ to $+0.200$; static ≈ 0.000 ; paired dist–conc full-FT deltas $+0.30$ to $+0.48$ dynamic, $+0.233$ static at $N=256$). At dynamic concentrated $N=16,384$ the full-fine-tuning deficit remains negative in every seed but is significant in only one of three. No LoRA cell shows significant degradation (worst cell -0.067 [$-0.167, 0.000$]); the 60 per-cell readouts carry no multiplicity adjustment and no noninferiority margin, so this is a failure to observe degradation, not an equivalence claim. The dynamic concentrated arm recovers exactly where $w_{\min}$ forces multiple worlds ($N=65,536$, 4 worlds: $+0.100$ in all seeds), tying the recovery to world count at fixed steps. In the K -indexed adaptation protocols, map count and state count move together; the factorial separates them: the protective variable is distinct-world coverage (or adapter-restricted capacity), not raw state count. Table 14 gives the direct distributed–concentrated contrast for every domain $\times N \times$ method cell: for full fine-tuning and scratch the interval excludes zero in all 10 cells with $w_{\min} \leq 2$ and in none of the 10 with $w_{\min} \geq 4$; LoRA’s contrasts are small throughout (two dynamic cells at $+0.117$ reach significance).

Protocol detail. Measured minimal world counts $w_{\min}(N)$ for $N=256/1,024/4,096/16,384/65,536$: static concentrated 2/6/22/89/356 worlds (distributed $8 \times$: 16/48/176/712/2,848); dynamic concentrated 1/1/1/1/4 (distributed 8/8/8/8/32). In the dynamic domain, rank-8 LoRA wraps every Conv2d weight of the field U-Net with a shape-agnostic adapter parametrization (zero-initialized B , so the adapter is the identity at start; 20 wrapped convolutions; 192,080 trainable A/B parameters, 4.9% of the 3.89M-parameter model, base frozen); the static domain uses the scalar HRM’s rank-8 linear adapters. Every adaptation runs exactly 2,560 optimizer steps—the epoch-equivalent of the largest N under the canonical recipe—so compute cannot proxy for labels; smaller- N cells cycle their fixed sample. The three adaptation seeds vary initialization and batch order only; the sampled state indices are shared across methods and seeds within a cell. *Degradation* denotes a cell whose adapted success falls significantly below the trained source model’s zero-shot success (paired map CI excluding zero in the harmful direction); we reserve *collapse* for the constant-prediction optimization failures of Sections E and I. Inference unit: evaluation maps, seeds averaged within maps.

Limitations. Euclid solves 1/20 dense dynamic test maps at the binding budget, so dynamic matched-ratio cells have $n=1$ and carry no map-level uncertainty; dynamic inference rests on the 20-map paired success deltas. One target family per domain; three adaptation seeds; the coverage factor is the number of distinct procedural worlds, not a measured

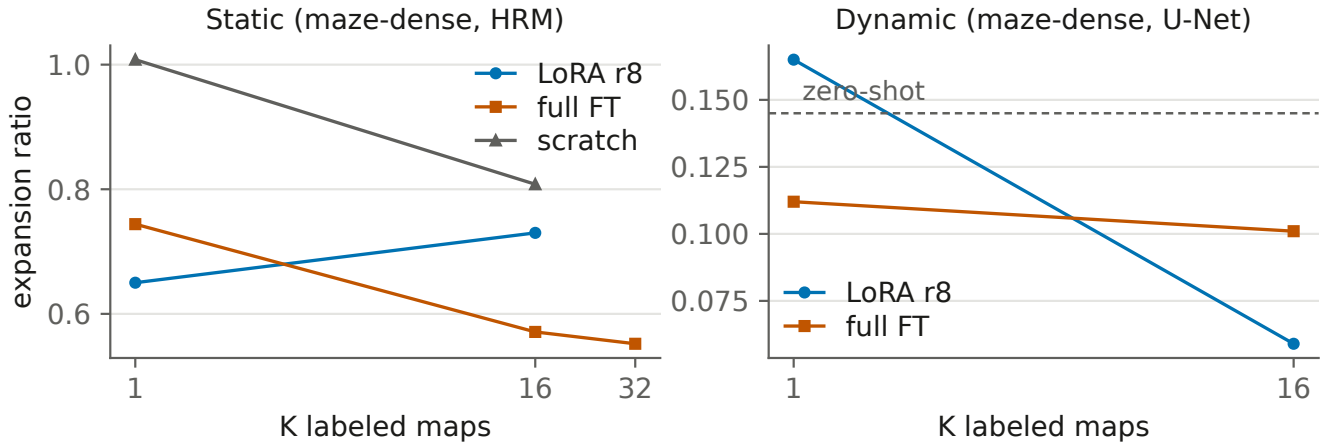


Figure 2: Illustrative K -indexed adaptation curves on the static (left) and dynamic (right) dense-maze targets (moved from the main paper; the factorial figure there supersedes them as the primary adaptation visual). On this static target, LoRA leads at $K=1$ and full fine-tuning overtakes it by $K=16$; on the dynamic target the low-supervision failure is not observed in the K -indexed protocol. Curves connect the evaluated K values for one backbone per panel; quoted inference is map-clustered.

Cell	K	$\Delta\text{succ. [CI]}$	$\Delta\text{ratio [CI]}$	n
dense/HRM	1	-.240 [-.360, -.133]	+.112 [+0.052, +.171]	10
dense/HRM	16	+.027 [-.020, +.100]	-.149 [-.179, -.117]	10
dense/ON-L.	1	-.047 [-.147, +.073]	-.162 [-.223, -.089]	10
dense/ON-L.	16	+.000 [-.027, +.033]	-.126 [-.171, -.083]	10
bug./HRM	1	-.467 [-.587, -.340]	+.225 [+0.089, +.352]	13
bug./HRM	16	-.020 [-.120, +.080]	-.145 [-.239, -.043]	16
bug./ON-L.	1	-.093 [-.207, +.007]	+.229 [+0.117, +.348]	15
bug./ON-L.	16	+.100 [+0.013, +.207]	-.052 [-.154, +.033]	15
rl./HRM	1	-.600 [-.773, -.407]	+.290 [+0.207, +.360]	11
rl./HRM	16	+.073 [-.027, +.187]	-.366 [-.489, -.235]	11
rl./ON-L.	1	-.293 [-.433, -.160]	+.013 [-.156, +.167]	11
rl./ON-L.	16	-.053 [-.167, +.053]	-.170 [-.326, -.010]	11

Table 13: Direct map-paired full-FT – LoRA contrasts (dense = dense maze, bug. = bugtrap, rl. = large rooms; ON-L. = ON-LSTM). Adaptation seeds are averaged within maps; intervals are 10k map-clustered bootstraps over the 30 test maps per target; Δratio is computed on the n triple-matched maps solved by both methods and the anchor, as a difference of expansion ratios relative to the anchor (negative favors full fine-tuning). At $K=1$ the success contrast favors LoRA in all six cells (four significantly); at $K=16$ the effort contrast favors full fine-tuning in all six cells (five significantly) at near-parity success.

geometric diversity; matching covers retained labels and optimizer steps, not the cost of generating and labeling additional worlds. Adaptations ran on cloud L4 hardware from locally collected, hash-recorded datasets.

Independent world-draw replicates (C14-R; frozen design 2026-07-25). Each original cell used one deterministically sampled world set; C14-R re-draws the world streams independently (draw offsets entering only the pool seeds; test cohorts, budgets, and sampling rules unchanged) at the decisive cells: dynamic $N \in \{1024, 16384\}$ and static $N=256$, both coverage levels, all three methods, one optimization seed, 18 arms per draw on the same frozen test cohorts. Results across both draws (deltas vs the frozen source, map-level CIs), reported as-is against the preregistered readouts. *The coverage contrasts replicate in every draw:* distributed–concentrated is positive in all 18 replicated contrasts, with CIs excluding zero for

scratch in all six (up to +0.500) and for full fine-tuning in five of six—draw 2 dynamic +0.350 [+0.150, +0.550] and +0.200 [+0.050, +0.400]; draw 3 dynamic +0.400 [+0.200, +0.600] and +0.300 [+0.100, +0.500], static +0.133 [+0.033, +0.267]; the one exception is draw 2’s static contrast, +0.067 [+0.000, +0.167], whose lower bound touches zero—and the distributed arms are rescued in every draw (R2 passes twice). *Concentrated harm is directionally consistent but power-limited for full fine-tuning:* scratch’s dynamic concentrated harm is significant in both draws (−0.300 [−0.550, −0.050] each), while full fine-tuning’s reproduces in sign and magnitude (−0.200 and −0.250 at $N=1024$) with CIs grazing zero at $n=20$ maps and one optimization seed, so strict R1 fails in both draws. *The harm’s magnitude depends on the drawn worlds:* draw 3’s static concentrated cells are significantly harmful for all three methods (full FT −0.233; scratch −0.500; LoRA

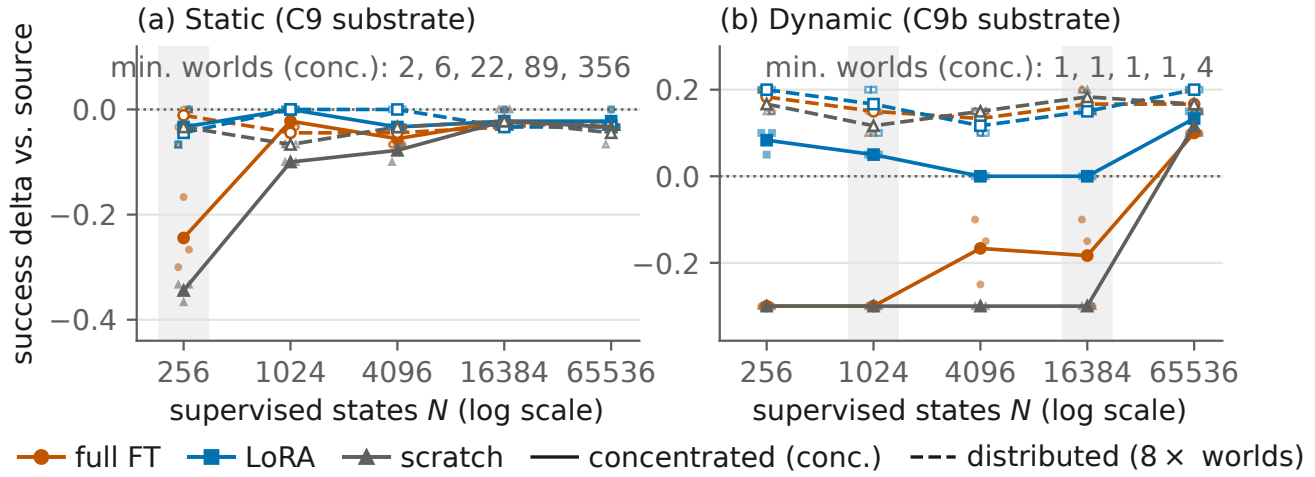


Figure 3: The full matched-label factorial (both domains; the main paper shows the dynamic panel). Success delta versus the frozen zero-shot source at fixed supervised-state count N on (a) the static and (b) the dynamic substrate. Solid: concentrated ($w_{\min}(N)$ worlds, annotated); dashed: distributed ($8\times$); small markers are the three adaptation seeds. Shaded bands mark the N levels replicated with independent world draws (C14-R).

Domain	N ($w_{\min}$)	Full FT	LoRA	Scratch
static	256 (2)	+ .233* [+ .111, + .378]	− .011 [− .067, + .044]	+ .311* [+ .156, + .467]
static	1024 (6)	− .022 [− .056, + .000]	.000 [.000, .000]	+ .033 [− .033, + .122]
static	4096 (22)	+ .011 [− .033, + .067]	+ .033 [+ .000, + .100]	+ .044 [+ .000, + .122]
static	16384 (89)	− .011 [− .033, + .000]	− .011 [− .033, + .000]	.000 [.000, .000]
static	65536 (356)	+ .011 [+ .000, + .033]	− .011 [− .033, + .000]	− .011 [− .033, + .000]
dynamic	256 (1)	+ .483* [+ .283, + .700]	+ .117 [+ .000, + .267]	+ .467* [+ .250, + .683]
dynamic	1024 (1)	+ .450* [+ .250, + .650]	+ .117* [+ .017, + .250]	+ .417* [+ .217, + .633]
dynamic	4096 (1)	+ .300* [+ .133, + .483]	+ .117* [+ .017, + .250]	+ .450* [+ .250, + .650]
dynamic	16384 (1)	+ .350* [+ .183, + .533]	+ .150 [+ .000, + .300]	+ .483* [+ .283, + .700]
dynamic	65536 (4)	+ .067 [+ .000, + .183]	+ .067 [+ .000, + .183]	+ .050 [+ .000, + .133]

Table 14: Direct distributed–concentrated success contrasts, all 30 factorial cells: paired per-map differences with adaptation seeds averaged within maps and 10k map-bootstrap 95% CIs (static $n=30$ maps, dynamic $n=20$). *: interval excludes zero (marginal; no multiplicity adjustment). For full fine-tuning and scratch the significant cells are exactly those where concentration forces at most two distinct worlds ($w_{\min}$ in parentheses).

−0.133 [−0.267, −0.033], failing R4—the first significant LoRA degradation in the program, still the least-degraded method in its cell), while draw 2’s static concentrated worlds happen to be benign (full FT −0.067, n.s., failing R3; LoRA never degrades in draw 2, passing R4). Synthesis: the coverage mechanism replicates under independent world draws; the concentrated-harm magnitude varies with which worlds are drawn; and LoRA’s protection is “most stable, not absolute”—significantly degraded in exactly one cell of one draw across the original factorial and both replicates. Raw rows for both draws ship in the artifact package.

Dom./cov.	N	Full FT	LoRA	Scratch
static conc	256	-.30--.17*	-.07-.00	-.37--.33*
static conc	1024	-.03-.00	.00	-.10
static conc	4096	-.07--.03	-.03	-.10--.07
static conc	16384	-.03-.00	-.03-.00	-.03-.00
static conc	65536	-.03	-.03-.00	-.03
static dist	256	-.03-.00	-.07--.03	-.07-.00
static dist	1024	-.07--.03	.00	-.07
static dist	4096	-.07--.03	.00	-.03
static dist	16384	-.03	-.03	-.03-.00
static dist	65536	-.03-.00	-.03	-.07--.03
dynamic conc	256	-.30*	+.05+-.10	-.30*
dynamic conc	1024	-.30*	+.05	-.30*
dynamic conc	4096	-.25--.10	.00	-.30*
dynamic conc	16384	-.30--.10 [†]	.00	-.30*
dynamic conc	65536	+.10	+.10+-.15	+.10+-.15
dynamic dist	256	+.15+-.20	+.20	+.15+-.20
dynamic dist	1024	+.15	+.10+-.20	+.10+-.15
dynamic dist	4096	+.10+-.15	+.10+-.15	+.15
dynamic dist	16384	+.15+-.20	+.15	+.15+-.20
dynamic dist	65536	+.15+-.20	+.20	+.15+-.20

Table 15: C14 per-cell success deltas versus the frozen source (range across the three adaptation seeds; a single value means all seeds agree). *: the paired map CI excludes zero in the harmful direction in every seed. [†]: significant in one of three seeds. Per-seed CIs for all 180 cells ship in the artifact package.

Term	Coefficient [95% CI]
intercept	+0.7546 [+0.6969, +0.8198]
$\log_2 N$	-0.0140 [-0.0190, -0.0081]
full FT	+0.0022 [-0.0691, +0.0842]
scratch	-0.0213 [-0.0402, -0.0050]
dynamic	-0.4387 [-0.4901, +0.0000]
distributed	+0.0012 [-0.0197, +0.0205]
full FT $\times \log_2 N$	-0.0025 [-0.0087, +0.0028]
full FT $\times$ dynamic	-0.0409 [-0.0585, +0.0000]

Table 16: C14 preregistered map-level regression of the matched expansion ratio on $\log_2 N$, method, domain, and coverage (map-clustered bootstrap CIs). H-C14 required the full-FT $\times \log_2 N$ interaction to be significantly negative; it is null.

Gate	Verdict
K=0 continuity	FAIL, diagnosed as false premise: C7 had no MLP arm and did not give every architecture the C11 input; real shallow separation plus collapsed recurrent cells violate the all-tie assumption. Measurement layer verified.
G1 dose-response	Negative: no structured arm beats MLP at ≥ 2 mission depths with non-decreasing gap.
G2a forced compute	Negative: HRM-v2 $k=1/2/4/8$ expansion and MAE curves flat in every $K=8$ config.
G2b learned halting	Negative, inverted: map-level Spearman $\rho = -0.578$, stratified permutation $p=5 \times 10^{-5}$, $n=225$ maps (record-level $\rho = -0.407$, $n=675$, the original gate value); mean would-halt steps 6.79/7.01/5.30 at $K=2/4/8$ (record level; map level 6.78/7.01/5.30).
G3 closure	Neither preregistered branch: global-input arms separate at shallow K (U-Net/GNN vs. MLP ratios $\approx 0.67-0.87$; state MAE 0.151/0.171 vs. 0.251 at $A/K=0$), but no arm converts depth into growing advantage.

Table 17: C11 preregistered gates.

E. C11 Gates, Collapse Forensics, and Scale

Collapse forensics. Five of 33 recurrent/ACT arm-cells show constant-prediction or partial seed collapse (HRM-trace A8/C8, ON-LSTM B0, HRM-v2 B0, two ON-LSTM A8 seeds), with a reproducible loss signature at the normalization cap. The largest recurrent deficits (HRM-trace ratios 1.526 at A8, 1.600 at C8) are collapse measurements, not expressiveness estimates. Removing $K=0$ padding *worsens* recurrent predictions: models used pad steps as extra recurrent compute.

Scaled addendum. A 12-run `unet_film_big` vs. `hrm_trace_big` comparison (complete; 2,700 eval rows) preserves the ordering: completion 0.804 vs. 0.736 at $K=2$ and 0.413 vs. 0.307 at $K=8$; mean state MAE 0.395 vs. 0.447 ($K=2$) and 0.906 vs. 1.855 ($K=8$). At $K=8$ the scaled HRM exactly matches the leg-sum baseline’s completion and mean expansions (0.307; 701.34)—persistent collapse, not recovered hierarchy.

F. C12 Detail

C12-A (memory headroom, then learned pilot): constructed decision-relevant aliasing at 71.3% of paired decisions; privileged mode/history diagnostic +0.467 completion and -65.1% collision-adjusted regret (map-bootstrap 95% CI 63.0–66.9%); oracle completion 0.975. The one-seed matched temporal-hierarchy pilot fails G1 (forecast), G2 (planning), and G3 (persistent carry); overall *strong negative*. C12-B (tied refiner): cycle-1 $\rightarrow$ 8 normalized burden improves monotonically in every cell ($A/K2$ 0.859 $\rightarrow$ 0.815; $A/K8$ 0.624 $\rightarrow$ 0.576; $C/K2$ 0.867 $\rightarrow$ 0.777;

Rung	Factor changed	Δ exp. [CI] / (wins)
C	fresh certifier (oracle rank)	+11.50 (0/6)
D	shared-queue certifier (oracle)	−6.83 (6/6)
E	exact rollout rank	+1.33 (2/6)
F	monotone recalibration	+1.33
G	analytic local escape	+6.83 (0/6)
I	one-family model, live 6 suites	+15.36 [+12.02, +18.64]
J	suite-balanced, static insert	+16.17 [+9.67, +22.67]
K	+ one local Bellman backup	−1.21 [−8.04, +5.46]
L	+ scale $\alpha=1.5$ (48 maps)	−13.04 [−18.67, −7.65]
M	frozen 144-map confirmation	−12.96 [−16.30, −9.74]

Table 18: The C13 falsification ladder. Target repairs (E, F), analytic constructions (G), and distribution repairs (I, J) fail; the integration repair (K) plus scale calibration (L) flip the sign, and the frozen confirmation (M) reproduces it.

$C/K8$ 0.576 $\rightarrow$ 0.517; all cycle-1/8 map-bootstrap intervals separate), but the $K8-K2$ gain is +0.0038 (CI [−0.0199, 0.0281]) on A and -0.0305 [−0.0726, 0.0096] on C—no dose-response. Tied beats shallow and untied on $C/K8$ (deltas 0.0177/0.0076, BH $q=6.7 \times 10^{-5}$) but loses to untied on $A/K8$ ($q=0.0168$). Bellman residual *worsens* over cycles in every cell while burden improves; solved cycle-8 paths average 1.4–2.2% above oracle cost. $K=16$ was dropped before training by a construction gate (only 2/300 and 1/300 valid missions).

G. C13: Mechanism Ladder and Frozen Confirmation

Claim supported. A provider restricted to bounded local observations (radius-0.20 subgraph on a known roadmap; behavior-derived labels; no shortest-path supervision) can, with one inference-time local Bellman backup, beat every fixed complete-map C7 provider in pooled expansions at matched empirical path quality on a frozen, disjoint cohort.

Ladder. Each rung changes one factor; deltas are paired expansions versus the matched comparator (rungs C–G versus matched focal search at $w=1.10$, including oracle providers on six development maps; rungs I–M versus the complete-map field HRM, live on all six suites).

Frozen confirmation, per suite. 144 maps (24 per family; seeds disjoint from every prior cohort; feature caches hash-verified against live regeneration; model, radius 0.20, $\alpha=1.50$, integration, and analysis frozen before generation).

Scale calibration. On the disjoint 48-map calibration block, $\alpha \in \{1.00, 1.25, 1.50, 2.00\}$ gives deltas -3.79 [−9.40, +1.73], -9.77 [−15.38, −4.42], -13.04 [−18.67, −7.65], and -16.94 [−22.56, −11.54], with 4/4/6/6 negative suite means and mean/max cost ratios rising from 1.022/1.241 to 1.052/1.366; $\alpha=1.50$ was frozen as the largest scale whose suite breadth was complete before the absolute-ceiling verdict, and every arm formally fails the preregistered absolute 1.10 maximum-cost ceiling. *Chronology, stated explicitly:* the absolute ceiling was part of the original calibration design; its failure was observed on the calibration block, during development and before any confirmation data existed; the comparator-relative criterion (mean

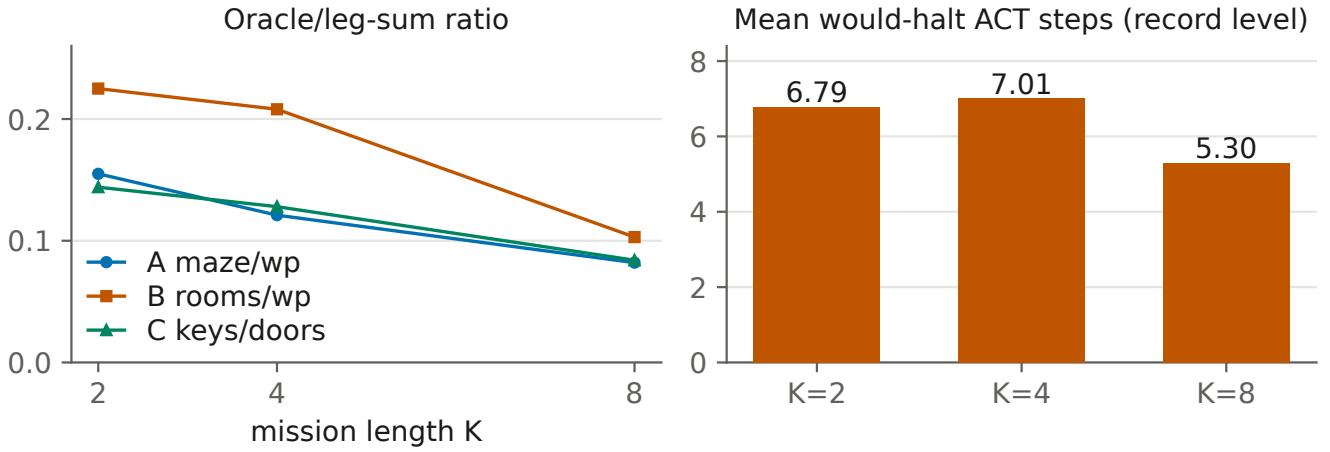


Figure 4: Compositional-mission hierarchy test (C11); K here is mission length. Left: the exact oracle’s ratio over a leg-sum baseline falls as missions deepen (headroom exists). Right: learned HRM halting uses *fewer* steps at $K=8$; bar labels are record-level means (map-level 6.78/7.01/5.30); both analysis grains: map-level $\rho = -0.578$, $p = 5 \times 10^{-5}$, $n=225$ maps; record-level $\rho = -0.407$, $n=675$.

Suite	Local	Field HRM	Δ	95% CI	W/T/L
Maze	47.63	84.38	-36.75	[-45.58, -27.83]	24/0/0
Dense maze	95.38	104.50	-9.13	[-15.21, -3.13]	18/1/5
Rooms	98.04	106.17	-8.13	[-14.33, -1.71]	16/0/8
Spiral	120.25	121.33	-1.08	[-5.88, +4.21]	14/1/9
Bugtrap	14.96	24.29	-9.33	[-17.79, -2.92]	18/1/5
Large rooms	33.58	46.92	-13.33	[-18.92, -7.50]	19/0/5
Pooled	68.31	81.26	-12.96	[-16.30, -9.74]	109/3/32

Table 19: C13-M per-suite paired expansion deltas versus the complete-map field-HRM comparator. Every suite mean is negative; the spiral interval crosses zero, and the preregistered gate is pooled with a suite-breadth condition (at least four negative suite means), which passes. The local provider also beats every other fixed C7 learned provider in pooled expansions on this cohort.

path-cost ratio within $+0.005$ and maximum within $+0.02$ of the complete-map field HRM) and the $\alpha=1.50$ selection were then written into the frozen confirmation design *before* the 144-map cohort was generated, and no criterion changed afterward. Under that gate the confirmation passes with margin (1.0235/1.1624 vs. 1.0311/1.3346); had the original absolute ceiling been retained, the direct arm would fail it on maximum cost, which we report rather than suppress.

Operating points and timing. The direct $\alpha=1.50$ arm is the matched-quality expansion result and carries no formal bound. The separate certified operating point runs the learned rank inside a $w=1.10$ focal band with a shared-queue consistent-bound incumbent proof and passes all 144 bound and certificate checks. Per-map wall time decomposes as feature construction 5.138 s (prototype Python builder), model inference 0.0008 s, local backup 0.0097 s, and search 0.0002 s, versus 0.371 s complete-map field-HRM inference: the result is search effort and path quality, not latency.

Architecture substitutions under the identical method (C13-N/O/P) are summarized in the main paper; their raw artifacts (development grids, verdicts, integrity manifests) ship in the artifact package.

H. Discrete Program Detail

The discrete program is larger than the canonical subset reported here (fifteen experiment families, including forecaster lineages, diffusion baselines, an empty-model baseline-rerun control, and a budget-invariance analysis showing that raising budgets adds work without rescuing success on the hard families); the paragraphs below report the completed canonical results, and the full inventory ships with the released artifacts.

Forecasting. Matched 100-episode protocol: HRM-3M/10M 71/100 vs. LSTM 67–69/100 (no uncertainty reported; two-episode matched gap). Structured Preset M+ (four 100-episode suites): ON-LSTM-10M 0.3875 mean success vs. HRM-10M 0.2650 and HRM-3M 0.200; ON-LSTM-3M also has the lowest rollout MSE (0.1257/1.1269 one-/five-step vs. 0.3973/2.6438 for HRM-3M).

Additive transfer (clean v3, 22 suites $\times$ 3 budgets). Manhattan 0.590 success / 126,707 mean expansions; HRM full-FT 0.585 / 128,013; ON-LSTM full-FT 0.564 / 127,228; HRM LoRA 0.554 / 140,852; ON-LSTM LoRA 0.514 / 145,282. Every learned arm $\leq$ matched baseline.

Pooled multitask. HRM pooled base 0.612 success (+2.11pp) / 122,184 expansions vs. Manhattan 0.591 / 126,630—the strongest completed discrete additive positive; only 1 of several specialists beats its matched four-suite baseline (+2.0pp; others -0.42 to -20.25 pp).

Focal ranking (local pilot, 3–8 seeds). Magnitude is scale-flat despite near-perfect ordering (correlation 0.987–0.994; predicted/true mean 0.73/0.42/0.27/0.19 at 64/128/192/256). At $w=1.0$: HRM A128-static ratio 0.85 with success 0.62 $\rightarrow$ 0.75; A192 0.94 (0.75 $\rightarrow$ 0.75); A128-dynamic 0.93 (0.62 $\rightarrow$ 0.62); ON-LSTM A128/A192 0.85 (0.75 $\rightarrow$ 0.75). At $w=1.05$, A128-dynamic improves further but A192 success regresses 0.75 $\rightarrow$ 0.62; $w=1.0$ is the evidence-safe operating point. Experts exactly match pooled bases in tested cells.

I. Validity Forensics

The frozen-halt artifact. An early discrete adaptive-computation run reported “74.60% halt accuracy” that equals 100 minus its exact accuracy—the halt logit was frozen (the training loop omitted the halt loss), so the metric measured a constant-false classifier. The loss path was repaired and verified live; we note the 100– x pattern as a red flag for adaptive-computation metrics, and no main-paper claim uses this run.

The cap-collapse signature. C5’s scalar HRM collapse sits at a loss value pinned by the residual cap (2.612 in that setup, with constant predictions at the cap). The same class of signature reappears in C11: collapsed cells share an *identical* final loss (1.1397, exactly the constant-cap loss under that normalization), enabling automatic collapse detection instead of misreading collapsed cells as capacity evidence.

Pad steps as compute. In C11, removing $K=0$ padding tokens *degraded* recurrent predictions—models had co-opted padding as additional recurrent iterations. Sequence-model comparisons that alter padding alter compute.

J. Statistics and Map-Clustered Reanalysis

Primary-test policy. Exact McNemar with BH correction is the primary success test within each experiment’s suite family; paired bootstrap intervals are descriptive companions, and where a marginal interval excludes zero but the corrected test does not (crossing, under the weighted- A^* control), the corrected test governs. Matched-ratio and path-quality intervals are marginal and uncorrected unless stated otherwise. Multiplicity families are declared per experiment: the six dynamic suites for each zero-shot or control comparison, the six target/backbone cells per endpoint and K for the direct method contrasts, and the six suites of the C13 confirmation.

Success is compared with paired McNemar tests, BH-corrected within stage families. Expansion effort is compared only on matched solved maps via median ratios with bootstrap percentile CIs (10k–20k resamples); from C12 onward, bootstraps are map-clustered by design. C9–C11 originally pooled repeated evaluations of fixed test maps across adaptation/model seeds; Section J reports the map-clustered reanalysis of every quantity the main paper quotes from those stages; the fixed-provider dynamic reanalysis of Table 7 uses the same map-level grain. Matched-solved ratios condition on joint success and can favor arms that solve only easier maps; we therefore always report success and ratios separately and never combine them into a single score. Negative results are failures to observe an effect under the tested protocol, not equivalence claims; no equivalence tests were run.

Map-clustered reanalysis

Units are test maps; adaptation/model seeds are averaged within maps before inference; success deltas use 10,000-resample map bootstraps; expansion ratios are medians of per-map ratios over matched maps at the binding budget in additive mode; the C11 halting test aggregates model seeds within maps and permutes mission length within config (maps are independent across K cells; 20,000 stratified permutations). Estimator conventions differ slightly from the published record-level curves, so point values differ modestly; the question is verdict-level agreement. A methodological caution surfaced during this work: a first pass that accidentally pooled additive and focal evaluation modes attenuated the apparent C9 transfer effects by roughly half—mode/aggregation mixing alone can manufacture or destroy an effect of the size under study.

C11 halting. Record level reproduces the published value ($\rho = -0.4066$, $n=675$); map level strengthens it ($\rho = -0.5778$, $n=225$, stratified permutation $p=5\times 10^{-5}$). Map-level means 6.78/7.01/5.30 at $K=2/4/8$.

C11 vs-MLP ratios. Map-clustered ratio-of-means CIs exclude parity at $K \in \{0, 2\}$ in config A for both U-Net (0.734/0.846) and GNN (0.822/0.923), and in config B at $K=0$ (0.904/0.818) plus B/ $K=2$ for the GNN (0.805); config C never separates; every $K \in \{4, 8\}$ cell includes parity.

C9H bound-vs-capacity. 27 cells: median bounded-minus-unbounded map-level ratio delta +0.0000; 15/27 exactly zero; sd 0.0077; max $|\Delta|$ 0.0370—the capacity attribution stands.

C9B aware—blind (full FT, $K=16$). Eight of nine point deltas are zero or negative; the single positive (+0.017

Cell (HRM)	Δ succ. [CI]	map ratio [CI]
dense LoRA $K=1$	+ .467 [+ .300, + .633]	.686 [.590, .774]
dense LoRA $K=16$	+ .433 [+ .267, + .600]	.786 [.694, .891]
dense full $K=1$	+ .227 [+ .080, + .380]	.789 [.751, .836]
dense full $K=16$	+ .460 [+ .287, + .633]	.610 [.549, .670]
dense scratch $K=1$	- .167 [- .300, - .047]	1.035 [1.029, 1.039]
rl. full $K=1$	- .167 [- .340, + .013]	1.293 [1.170, 1.503]
rl. full $K=8$	+ .387 [+ .227, + .547]	.590 [.498, .712]
bug. LoRA $K=32$	- .320 [- .507, - .133]	1.153 [1.012, 1.523]

Table 20: Map-clustered C9 key cells (binding budgets, additive mode; dense = maze-dense, rl. = rooms-large, bug. = bugtrap). The $K=16$ full-FT/LoRA CIs are disjoint (the crossover), the rooms-large $K=1$ catastrophe excludes parity, and the bugtrap LoRA degradation is significant in both metrics.

[0.000, +0.050]) does not exclude zero; no cell shows a significant aware advantage.

C10 composition. Map-paired deltas vs. zero-shot: four cells ≈ 0 with CIs crossing zero; the two ON-LSTM rooms cells are significantly *worse* (+0.089 [+0.061, +0.121] and +0.098 [+0.076, +0.123]), matching the published point deltas and upgrading the null to “never better, sometimes significantly worse.”

No preregistered verdict flips under any of these recomputations.

K. Reproducibility and Artifact Inventory

From C7 onward every stage freezes a written design (gates, budgets, analysis grain) before execution; artifacts (checkpoints, feature caches, cohort manifests, raw rows, verdicts, reports) are hash-bound in integrity manifests; confirmation cohorts are generated only after freezing model, integration, and evaluation code; failed development gates block confirmation rather than triggering retuning. Duplicate-evaluation determinism is enforced where preregistered (byte-identical traces; identical decision rows). The final-stage pilot on persistent planning-time state ran under a fully test-verified harness (158 focused-plus-neighboring tests; hardware pre-flight). Its first staged execution ended at a preregistered execution-integrity hard stop—the harness’s independent reanalysis disagreed with the primary comparison computation, voiding all mechanism fields. After repair of the named defect only, a fresh-fingerprint rerun passed every integrity primitive, including independent reconstruction of the comparison evidence from promoted raw artifacts, and returned the valid negative reported in the main paper: persistent carry loses to the same-checkpoint reset twin on map-macro frontier ranking (MRR delta -0.029 , 95% CI $[-0.060, -0.003]$), and no self-bootstrap or confirmation was run. We note the initial stop as the integrity machinery functioning as designed.

The anonymized artifact archive submitted with this paper contains: (i) the analysis scripts that regenerate the quoted map-level statistics from raw rows (the fixed-provider dynamic reanalysis, the map-clustered reanalysis, the weighted-A* analysis with its exact McNemar, ratio-CI,

and matched path-quality computations, the budget-curve derivation, the reachable-label recount, the factorial analysis, and the MovingAI external-benchmark analysis, plus all figure generators); (ii) the raw paired evaluation rows behind every quoted quantity (C7/C8 evaluation tables, including the fresh-cohort and both retrained-pipeline replication rows, the weighted-A* development and confirmation rows with the tuned-weight manifest and the extended-grid development/confirmation rows with their selection report, the SIPP baseline rows with per-instance gate checks and timings, the MovingAI external rows with the map manifest, the wall-time decomposition sample, C9/C9H/C9B/C10 evaluation tables, C11 evaluation and halting tables, C13 development and confirmation rows, the C14 per-cell outputs with sampled-index manifests, the measured $w_{\min}$ stream, and the 30-cell coverage-contrast outputs, and the C14-R independent-draw replicates); (iii) map/cohort manifests, seed lists, frozen configurations, the binding-budget calibration file, and the integrity manifests for the frozen confirmations; (iv) the preregistration design documents from C7 onward, including the C14 design with its pre-execution amendment and post-completion supersession record, the 2026-07-25 SIPP, MovingAI, and C14-R designs, and the 2026-07-26 extended-grid design with its disclosed chronology; (v) the two frozen source checkpoints the headline claims rest on (the fixed dynamic provider and the frozen static adaptation source), with all other checkpoints regenerable from the recorded commands and seeds; and (vi) a top-level README with reproduction recipes and expected outputs (the README states explicitly that analysis-script path constants must be pointed at the package’s raw directories when running standalone).

References

Sturtevant, N. R. 2012. Benchmarks for Grid-Based Pathfinding. *IEEE Transactions on Computational Intelligence and AI in Games*, 4(2): 144–148.